

Resume

Aarif Khan

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Experience

Software Engineer, Tekion Corp. — Bengaluru, India — Jan 2025–Present (joined as Software Engineering Intern)

- Designed and delivered modular, plug-and-play webhook infrastructure processing 500 Kafka records/sec, enabling new publishers to onboard by adding a single class — no changes to existing code
- Reduced p99 latency from 400ms to under 200ms (a 50% improvement) by root-causing and resolving a p99 latency SLA breach on a production open API GET endpoint through query optimization.
- Decreased average latency from 350ms to 10ms (a 97% improvement) on a read-heavy config API handling 50 req/sec by caching infrequently-changed config values in memory.
- Owned backend design and implementation for a GM vehicle check-in feature end to end, building a screen-agnostic job-creation API that consolidated 4 separate screen integrations into a single reusable contract with no per-screen logic
- Built a Systems Integrator platform enabling third-party companies to develop integrations on Tekion's open APIs — deployed in an isolated cluster for security, reducing integration go-to-market time by shifting delivery from a single internal team to multiple third-party teams working in parallel, validated by delivering the first proof-of-concept integration with a small team
- Diagnosed and fixed a P0 production incident where a warranty posting defect caused ~2,000 invalid entries for the first dealer migrated to a new system (out of 2,000 total customers);

Associate Software Developer, Location Solutions, LLC — Dubai (Remote, Part-time) — Mar 2024–Aug 2024

- Rewrote a legacy .NET/C# service in Node.js, improving performance, scalability, and maintainability; reduced average response time by 20% for a specific workload, used by every Location Solutions customer
- Implemented in-memory join algorithms for time-series application data, avoiding repeated single-record database read/write round-trips of the database.
- Increased backend algorithm test coverage to 60% through comprehensive unit testing, improving reliability and reducing regressions
- Integrated OpenTelemetry logging into a legacy service, enabling retroactive debugging of customer-reported failures that previously had no visibility.

Projects

Real Time Object Detection on a Video Stream from Drone/Mobile — Team/hackathon project, built during CodeUtsava 7.0 —

[Source](#)

- Problem solved: pothole detection using drone footage and machine learning.
- Tools & technologies used: MERN stack, Flask, OpenCV, YOLOv8, RTMP protocol.
- Took an input stream from an RTMP source and ran object detection on the video frames.
- Built the detection script with OpenCV to run object detection and send frames to the server and database.
- Built an API to store detected frames into the file system and MongoDB database.
- Built a frontend UI in React to interact with the detected objects.

Intelligent Haze Classification and Dehazing — Independent project, Bachelor's minor project — [Source](#)

- Tools & technologies used: Python, TensorFlow/Keras, CNNs, Res2Net, PixelShuffle.
- Built a 3-class CNN classifier (No Haze / Mild / Dense) to route hazy images to a haze-level-appropriate dehazing network, reaching 93% validation accuracy, up from 37% at epoch one.
- Implemented a teacher-student knowledge-transfer dehazing network (based on Wu et al.'s KTDN architecture) from a research paper with unspecified training details, combining L2, SSIM, and VGG19 perceptual losses for the teacher network.
- Trained the teacher network on a combined RESIDE/I-Haze/O-Haze/NH-Haze dataset, reducing validation loss from 0.40 to 0.15 over 200 epochs.

Video Conferencing Website — [Live](#) · [Source](#)

- Tools & technologies used: React.js, Express.js, Socket.io, WebRTC (Peer.js).
- Users create unique code-identified meeting rooms and share them to invite others.
- Built the backend server with Express.js to handle rooms and connection setup, using peer-to-peer WebRTC for video streaming and Socket.io for WebRTC signalling.

Education

B.Tech in Computer Science and Engineering — National Institute of Technology, Raipur — 2025 CGPA/Percentage: 7.52/10

Skills

Languages: C++, JavaScript, Java

Developer Tools: Git, Postman, Chrome Developer Tools

Frameworks: Spring Boot, Express.js, Node.js

Cloud / Databases: SQL, MongoDB, Kafka, Redis, Elasticsearch

Soft Skills: Communication, Team Collaboration, Problem-solving, Analytical Thinking

Areas of Interest: Operating Systems, Databases, Computer Networks

Achievements

- **Winner, CodeUtsava 7.0** — Central India's largest hackathon, organized by NIT Raipur — Nov 2023
- **Winner, CSE-MCA-IT branch, Vigyan Science Exhibition** — organized by NIT Raipur — Oct 2023
- **Codeforces Specialist** — Global Rank 530 in Codeforces Round 944 (Div. 4) — May 2024
- **Global Rank 2446/29648** in LeetCode Biweekly Contest 112